

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Python backend engineer with production experience building data processing pipelines and REST APIs at scale. Designed a Kafka-driven indexing pipeline for large-scale corpus ingestion (VectorLift), built an interactive 500GB scientific data platform serving 100+ researchers on GCP (NC State GRA), and shipped 10+ backend REST API features with S3 file storage and third-party API integrations at Isomer AI. Strong in AWS, PostgreSQL, MySQL, Docker, and cloud data workflows. MS CS at NC State (May 2026). Portfolio: kulkarniaditya.com

CORE COMPETENCIES

Python + FastAPI

Data Pipelines + Kafka

REST APIs

AWS + S3

PostgreSQL + MySQL

Docker + CI/CD

Prometheus + Grafana

React + Next.js

WORK EXPERIENCE

NC State University Apr 2025 - Present

Graduate Research Assistant

- Built and deployed a cloud-native scientific data platform on GCP Cloud Run serving **500GB of genomic datasets** to 100+ researchers worldwide; designed MERN stack backend with REST APIs for interactive data exploration and Grafana dashboards for performance monitoring
- Applied Python-based hyperparameter tuning and time-series ML analysis improving model accuracy by 12%; maintained reliable data pipelines supporting continuous research workloads

Isomer AI May 2025 - Present

Product Engineer Intern

- Shipped 10+ production backend REST API features in TypeScript/Node.js: S3 file storage pipelines, document processing (PDF generation, email parsing via Microsoft Graph API), multi-provider API integrations across 3 LLM providers and 8+ models -- **394 commits, 7 production deployments**
- Implemented Auth0 OAuth/RBAC, Temporal.io workflow orchestration for fault-tolerant data processing, PostgreSQL + Redis data layers with Prisma ORM, and Docker containerization with GitHub Actions CI/CD

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ Python AI-driven workflows for health data automation (OCR, NLP, ML pipelines), improving claim accuracy by 25%; automated AWS and GCP deployments with Terraform and integrated Grafana monitoring dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built Python distributed data pipelines using Hadoop + Spark for batch processing and Kafka streaming for real-time data ingestion, reducing ticket-processing latency by 20%; managed end-to-end SDLC for a large-scale data system

PROJECTS

VectorLift -- Semantic Search and Ranking Engine Python + Kafka + FastAPI

Built a production Python data platform: Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant vector store; BM25 + dense bi-encoder + cross-encoder re-ranker search pipeline evaluated across 6 configurations on MS MARCO (NDCG@10, MRR@10, MAP). Served via FastAPI REST API with per-stage latency tracking in Prometheus + Grafana. Full Docker Compose orchestration with PostgreSQL + Redis data stores.
Stack: Python, FastAPI, PyTorch, Kafka, Elasticsearch, Qdrant, PostgreSQL, Redis, Prometheus, Grafana, Docker Compose

LLM Evaluation Harness Python + Data Analytics

Built a vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types on a 2,000-example frozen test set with 3 repeated runs per model. Tracked ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, and cost/call; statistical significance via paired bootstrap CIs and McNemar test. Data stored in DuckDB/Parquet; results visualized in Streamlit.

Stack: Python, DuckDB, Parquet, Streamlit, Docker, Anthropic/OpenAI/Google GenAI SDKs

PulseRank -- Real-Time Leaderboard and Activity Feed MySQL + REST APIs + React

Full-stack data platform: NestJS REST API backend with MySQL transactional writes (idempotency-key deduplication, outbox pattern), MongoDB feed projections with TTL expiry, Redis sorted sets for sub-millisecond reads, and Server-Sent Events for live updates. Next.js frontend with data visualization. Testcontainers integration tests; k6 load benchmarking; OpenTelemetry + Prometheus observability.

Stack: TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

AlgoPulse -- Agentic AI Platform Event-Driven + Data Pipelines

Architected an event-driven data orchestration platform; re-engineered from API-polling to streaming Edge Functions + Temporal.io + Kafka for fault-tolerant state synchronization -- 2,000 async requests per second without data loss. React frontend with real-time data streaming.

Stack: TypeScript, Node.js, React, Temporal.io, Kafka, Supabase Edge Functions, Docker

EDUCATION

North Carolina State University -- Master of Science, Computer Science Aug 2024 - May 2026

Courses: AI, Data Science, Algorithms, Operating Systems, Software Engineering, Object Oriented Development

Mumbai University (DJSCE) -- Bachelor of Technology, Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | IEEE-published NLP research (T5 transformer, 1,500+ document pipeline); HackNC24 Top 10 of 100+ teams

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C Backend: FastAPI, Node.js, NestJS, Express, REST APIs, Temporal.io, Kafka

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DuckDB, SQLite

Cloud + Infra: AWS (S3, EC2), GCP (Cloud Run), Docker, Kubernetes, Terraform, GitHub Actions

Data + Analytics: Pandas, NumPy, Parquet, Streamlit, Prometheus, Grafana, OpenTelemetry Frontend: React, Next.js, Tailwind CSS

AI/ML: PyTorch, HuggingFace, sentence-transformers, FAISS, Qdrant, OpenAI/Anthropic/Gemini APIs, RAG